Semester II Final Exam **Bartonian ID _____**

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1) What is the molarity of 174.26g of K_2SO_4 dissolved in 500mL of solution?

1

2) What mass (in grams) of NaOH would be needed to make 500.0mL of 3.00M solution?

2

3) 500mL of 2.00M HCl is added to 500mL of distilled water. What is the molarity of the final solution?

3

4) Aqueous solutions of potassium iodide and lead (II) nitrate are mixed. Write the balanced, net ionic equation for this reaction. HINT: Alkali (group 1) metals and nitrates are universally soluble.

4

5) Which solution in each set is the better conductor of electricity (electrolyte)? Justify your answer.

a. 1.0M NaI or 1.0M $CaCl_2$

5a

b. 1.0M K_2CO_3, 1.0M PCl_3, or 1.0M NaBr

5b

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6) Use the chemical equation below for the combustion of methane to answer 6a and 6b.

$$\mathbf{CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)} \qquad \mathbf{\Delta H_{rxn} = -802.3\ kJ}$$

a. Does the forward reaction release or absorb heat?

6a

b. How much heat is released or absorbed when 64.2 g of methane gas reacts with excess oxygen?

6b

7) Use the chemical equation and the table of average bond energies to answer 7a and 7b.

$$\mathbf{CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O}$$

a. Write Lewis structures for all four species represented in the reaction above.

7a

CH_4 O_2 CO_2 H_2O

b. Use the *bond enthalpy method* to calculate the enthalpy change, ΔH_{rxn}, that occurs in this reaction.

7b

Bond	Average Bond Energies
O – C	351 kJ/mol
O = C	799 kJ/mol
O – O	142 kJ/mol
O = O	499 kJ/mol
C – C	348 kJ/mol
C = C	612 kJ/mol
C ≡ C	960 kJ.mol
H – H	436 kJ/mol
H – C	412 kJ/mol
H – O	467 kJ/mol
N – N	193 kJ/mol
N = N	418 kJ/mol
N ≡ N	941 kJ.mol
N – H	393 kJ/mol

8) A 3.60g sample of $CaCl_2$ was dissolved in a coffee cup calorimeter containing 100mL of distilled water. The temperature of the water increased from 21.5 °C to 27.5 °C during the dissolving process.

a. Find the amount of heat lost or gained by the water.

8a

b. Find the heat of reaction (ΔH_{rxn}) when one mole of calcium chloride dissolves in water.

8b

9) Determine the value of the enthalpy change, ΔH_{rxn}, for the following reaction using Hess's Law and the information provided below.

$C_2H_4(g) + 6F_2(g) \rightarrow 2CF_4(g) + 4HF(g)$

$2C_2H_4(g) \rightarrow 4H_2(g) + 4C(s)$ $\Delta H = -104.6$ kJ
$CF_4(g) \rightarrow C(s) + 2F_2(g)$ $\Delta H = +680.0$ kJ
$H_2(g) + F_2(g) \rightarrow 2HF(g)$ $\Delta H = -537$ kJ

9

10) Consider the following system:

$CaCO_3(s) + 2HCl(aq) \rightleftharpoons CaCl_2(aq) + H_2O(l) + CO_2(g)$

At a certain temperature, the equilibrium concentrations were found to be:

$[HCl] = 0.040M$ **$[CaCl_2] = 0.120M$** **$[CO_2] = 0.050M$**

a. Write the equilibrium expression, K_c, for this reaction.

10a

b. Calculate the equilibrium constant, K_c, for the reaction.

10b

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11) In which direction will the reaction below shift when additional $CO_2(g)$ is added to the equilibrium system?

$$\mathbf{CaCO_3(s) + CO_2(g) + H_2O(l) \rightleftharpoons Ca^{2+}(aq) + 2HCO_3^-(aq)}$$

11

12) Consider the following system at equilibrium. In which direction will the equilibrium shift if the pressure is reduced?

$$\mathbf{PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)}$$

12

13) In which direction will the equilibrium below shift when heat is added to the system?

$$\mathbf{N_2O_4(g) \rightleftharpoons 2NO_2(g)} \qquad \boldsymbol{\Delta H_{rxn}} \mathbf{= +57.2\ kJ}$$

13

14) In order to maximize the yield of hydrogen gas from coal, a plant supervisor suggested decreasing the temperature of the reactor to save on industrial heating costs. Would this produce more hydrogen gas? Briefly explain using the system below.

$$\mathbf{C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)} \qquad \boldsymbol{\Delta H_{rxn}} \mathbf{= +131.3\ kJ}$$

14

15) The total pressure inside a 5.0 L cylinder kept at 20°C is 1.5 atm. The pressure inside the cylinder increases to 2.5 atm. If the temperature remains the same, what is the new volume of the cylinder?

15

16) What mass in grams of $H_2O(g)$ can be produced at 27°C and 1.50atm when 32.84L of $H_2(g)$ react with excess $O_2(g)$?

$$\mathbf{2H_2(g) + O_2(g) \rightarrow 2H_2O(g)}$$

16

17) A 25.0mL NaOH(aq) sample of unknown concentration was titrated with 0.200M HCl(aq). When exactly 12.5mL of the titrant had been delivered, the analyte, which had been *pink*, turned *clear* and *colorless*. What was the original concentration, in mol/L, of this NaOH(aq) sample?

$$\mathbf{HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)}$$

17